# Pratik Nanaso Raut

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## PROFESSIONAL SUMMARY

MCA candidate with strong foundation in Spring Boot, REST APIs, and Object-Oriented Programming (OOP) — capable of building scalable backend systems.

Hands-on experience developing ERP application modules using Java and Spring Boot in Agile sprint workflows with focus on Data Structures & Algorithms (DSA) and Database Optimization.

## TECHNICAL SKILLS

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| **Languages** | Java, SQL, JavaScript, C++, C, Python (basics) |
| **Backend** | Spring Boot, REST APIs, Spring Security, Hibernate, JWT, JDBC |
| **Databases** | MySQL, MongoDB |
| **Tools** | Git, GitHub, Maven, Postman, IntelliJ IDEA |
| **Concepts** | Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), REST API Design, SDLC, Database Optimization, JWT Authentication, RBAC, Multithreading |
| **Exposure** | Cloud computing basics (AWS), AI/ML fundamentals, Big Data concepts |

## PROFESSIONAL EXPERIENCE

**Software Developer Intern** *Feb 2026 -- Ongoing*

*SetTribe IT Solutions | Pune*

- Developed ERP application modules using Java and Spring Boot, implementing RESTful API contracts for scalable service architecture.
- Operated in Agile sprint workflow using Git for version control, code reviews, sprint planning, and technical documentation.
- Designed and tested application modules with systematic debugging, reducing defect reopen rate by 40%.

## ACADEMIC PROJECTS

**PayStride** -- Multi-Tenant SaaS Payroll Management System *Java, Spring Boot, MySQL, Hibernate, REST API, JWT*

- Architected multi-tenant RBAC system using Spring Security and JWT, enforcing complete data isolation across all API endpoints.
- Optimized MySQL query execution through strategic indexing and schema normalization, reducing average response latency by 40%.
- Engineered BigDecimal payroll computation engine handling statutory deductions (PF 12%, ESI 0.75%) and 1.5x overtime multipliers with zero precision loss.

**EduTrack** -- Student Management Portal *Java, Spring Boot, MongoDB, REST API*

- Developed RESTful CRUD APIs over Spring Boot handling 500+ student records with optimized MongoDB aggregation queries.
- Designed modular service architecture enabling independent scaling of student, batch, and reporting modules.
- Implemented dual-layer validation (client-side and server-side) reducing data entry errors by 25%.

**Customised Virtual File System (CVFS)** -- In-Memory Linux-like File Manager *C, C++, Data Structures, OS Concepts*

- Simulated Linux file system operations (open, read, write, lseek) in C++ using inode-based metadata management and a custom shell interpreter.
- Achieved O(1) file lookup by designing hash table-based inode indexing with chaining for collision resolution.
- Implemented in-memory file allocation table using C++ classes managing file permissions, ownership, and hierarchical directory traversal.

## TECHNICAL STRENGTHS

- Strong foundation in Data Structures, Algorithms, and Object-Oriented Design.
- Proficient in backend application development, REST API design, and database-driven systems.
- Experience in Agile development workflows, Git-based version control, and collaborative engineering.
- Strong debugging, query optimization, and software architecture capabilities.

## EDUCATION

**Master of Computer Applications (MCA)** *2026 | Pursuing*

Dnyansagar Institute of Management and Research, Pune (SPPU) 8.25 CGPA | Sem VII: 7.24 SGPA

**Bachelor of Computer Applications (BCA)** *2024*

C.T. Bora College, Shirur (SPPU) 73.34%

**Higher Secondary Certificate (HSC)** *2021*

Maharashtra State Board 75%